PRACTICAL NO 06

Aim: Write a program to implement queue using linked lists.

Theory:

Write down about Queue

ALGORITHM:

Algorithm for Queue using Linked List

Step 1: Define Node Structure

- Each node has two parts:
    1. **data** → to store the value
    2. **next** → pointer to the next node

Maintain two pointers:

- **front** → points to the first node
- **rear** → points to the last node

Step 2: Enqueue (Insert element at rear)

1. Create a new node with given value.
2. If **rear == NULL** (queue is empty):
    - Set front = rear = newNode.
3. Else:
    - Link rear->next = newNode.
    - Update rear = newNode.
4. Print "value inserted".

Step 3: Dequeue (Remove element from front)

1. If **front == NULL**:
    - Queue is empty → print "Queue empty".
2. Else:
    - Store temp = front.

- Print "front->data removed".
- Move front to front->next.
- If front == NULL, set rear = NULL.
- Free temp node.

Step 4: Display Queue

1. If **front == NULL** → print "Queue empty".
2. Else:
   - Start from front.
   - Traverse nodes until NULL.
   - Print each node's data.

Step 5: Main Function

1. Call enqueue(10), enqueue(20), enqueue(30).
2. Call display() → shows all elements.
3. Call dequeue() → removes one element.
4. Call display() again.

Program:

```
#include <stdio.h>
#include <stdlib.h>

// Node structure
struct Node {
   int data;
   struct Node* next;
};

struct Node *front = NULL, *rear = NULL;
```

```
// Enqueue (insert)
void enqueue(int value) {
    struct Node* newNode = (struct Node*)malloc(sizeof(struct Node));
    newNode->data = value;
    newNode->next = NULL;

    if (rear == NULL) { // if queue empty
        front = rear = newNode;
    } else {
        rear->next = newNode;
        rear = newNode;
    }
    printf("%d inserted\n", value);
}

// Dequeue (remove)
void dequeue() {
    if (front == NULL) {
        printf("Queue empty\n");
        return;
    }
    struct Node* temp = front;
    printf("%d removed\n", front->data);
    front = front->next;
    if (front == NULL) rear = NULL; // queue empty
    free(temp);
}

// Display
void display() {
```

```
    struct Node* temp = front;
    if (temp == NULL) {
        printf("Queue empty\n");
        return;
    }
    printf("Queue: ");
    while (temp != NULL) {
        printf("%d ", temp->data);
        temp = temp->next;
    }
    printf("\n");
}

// Main
int main() {
    enqueue(10);
    enqueue(20);
    enqueue(30);
    display();

    dequeue();
    display();

    return 0;
}
```

OUTPUT:

```
10 inserted
20 inserted
30 inserted
Queue: 10 20 30
```

10 removed

Queue: 20 30